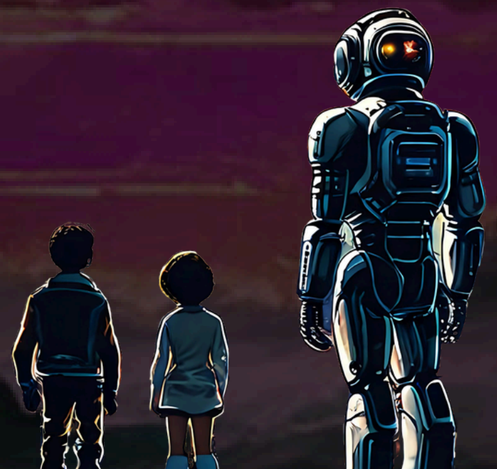
INDIA INNOVATES 2026

Team Name:

ALPHA-SEEKER

Members Name and Affiliation:



OMNIWARD AI
TEAM MEMBERS: DIVYANSH RAI, SAHIL CHANDRA, ABHINAV VERMA
AFFILIATION: RNS INSTITUTE OF TECHNOLOGY

PROBLEM STATEMENT

- *THE CHALLENGE:**
PROGRESS ON THE GROUND ISN'T BEING CLEARLY SHOWN TO THE CITIZENS WHO BENEFIT FROM IT.

- *THE GAP:**
MASS NARRATIVES OVERSHADOW LOCAL WINS, LEAVING REPAIRED ROADS, ADDED LIGHTS, AND SIMILAR UPGRADES UNSEEN. THIS CREATES NO REAL ACCOUNTABILITY OR CLEAR PROOF.

SOLUTION

- *Our Approach:**

OmniWard AI bridges the trust gap using computer vision and generative media. We ingest visual data of completed civic work, use AI to verify the physical improvement, and automatically render personalized "Before & After" update videos.

- *The Impact:**

This verified visual proof is geo-fenced and pushed directly via WhatsApp only to the voters registered in that exact micro-location, turning static administrative data into a hyper-local, transparent communication engine.

ARCHITECTURE

System Design:

1. **Data Ingestion**: Ground workers upload photos and videos showcasing the completed work.
2. **AI Verification Layer**: Using computer vision, the system validates the "After" state by comparing it against the "Before" state to ensure accuracy.
3. **Generative Media Engine**: The verified images are seamlessly integrated into a sleek, branded video graphic through an automated process.
4. **Targeting Engine**: A voter knowledge graph is utilized to link the project’s pin code with specific phone numbers for targeted communication.

*Integrations**: The vision model activates the rendering engine, which subsequently interacts with the WhatsApp Business API to distribute the

TECHNOLOGY USED

Tools & Frameworks: Python (Backend logic), OpenCV/YOLO (Computer Vision verification), C++ (For high-performance data processing if scaling), FFmpeg (Automated video rendering), Next.js (Admin Dashboard).

Contribution: Python and OpenCV handle the complex AI validation seamlessly, ensuring the system's efficiency and accuracy. C++ is leveraged for high-performance data processing, providing scalability when needed. FFmpeg automates video rendering, producing polished results. Next.js serves as the backbone for the admin dashboard, offering an intuitive interface for management. Our custom rendering pipeline ensures the output looks like a professionally edited campaign reel without any human intervention.

FEATURE / USP

Highlight the core features that make your solution unique. Emphasize its user experience, technical edge, or innovative elements that differentiate it from existing alternatives.

- *Cinematic Auto-Rendering:** Our solution goes beyond basic text alerts by providing high-retention visual proof, enhancing user engagement and understanding.
- *Tamper-Proof Verification:** Utilizing AI, we ensure the authenticity of physical work, effectively eliminating fraudulent contractor reports.

*Hyper-Local Geo-Fencing:** Our precision engine delivers updates about infrastructure developments tailored specifically to each user's immediate neighborhood, ensuring relevance and accuracy.

REFERENCES / LINKS

Include resources, research papers, APIs, or repositories that supported your project. You can also link to your GitHub repo, demo video, or documentation for further details. Resources: OpenCV documentation, Twilio/WhatsApp API architecture, National Voter Service Portal (NVSP) data structuring guidelines. For more information and to view all the videos and the mock project submitted, visit the GitHub repository at: [OmniWard-AI-GitHub]: (https://github.com/divyanshrai1/OmniWard-AI).

THANK YOU